

Physical Security Control Plan

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STEP 1

As a member of a cybersecurity team, you work for a corporation bidding on a government contract. All confidential information must remain onsite. Therefore, the contract prohibits saving confidential information to USB flash drives or other removable storage devices usable for transporting files offsite.

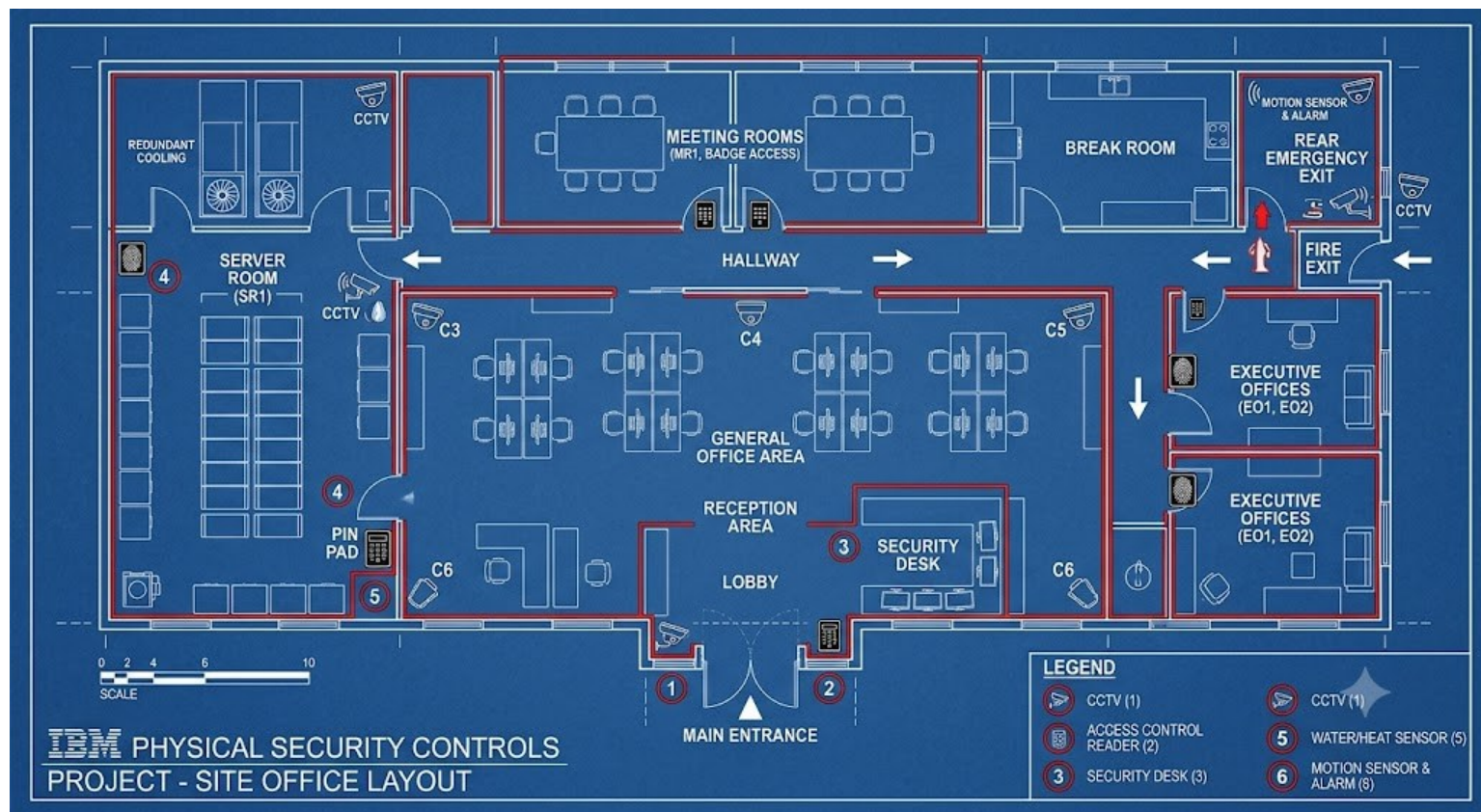
The corporation must improve its security measures to prevent this type of data breach.

Currently, the corporation has the following security measures:

- The server room is locked. Each IT staff member has a key to the server room.
- The door to the shared workspaces is not locked. However, a sign outside this door clearly states, "Authorized personnel only", and a receptionist screens all visitors.
- All employees have their own usernames and passwords.
- All employees have been instructed not to plug USB flash drives or external hard disk drives into their work computers.

Your supervisor tasks you with developing a plan for meeting the contract's requirements.

Insert the picture of your plan below.



New Security Measures

1. Electronic Access Control System (Key Cards)

- Replace physical keys with key card access.
- Restrict access to authorized personnel only.
- Maintain access logs for auditing.

2. CCTV Surveillance Cameras

- Install cameras at entrances, exits, workspaces, and server rooms.
- Monitor activities and investigate incidents.

3. USB Port Control

- Disable USB storage access through Group Policy and endpoint security software.
- Prevent confidential data from being copied to removable devices.

4. Visitor Management System

- Require visitor registration and temporary access badges.
- Escort visitors at all times.

5. Biometric Access to Server Room

- Use fingerprint or facial recognition for sensitive areas.

Physical & Infrastructure Controls for On-Premise AI Models & GPU Hardware

1. Physical Isolation for AI Model Infrastructure (ISO 42001 / NIST AI RMF):

Control: Mandate dedicated, air-gapped server racks for hardware running local Large Language Models (LLMs) or private AI agents to isolate them from standard corporate networks.

STEP 2

Identify at least three new physical controls to implement on the first floor.
Use the controls and placements that you chose in Step 1.

Explain your reasoning for each control, including its position.

Physical Controls:

1. Biometric Access Control at Server Room Entrance

Reasoning:

- Prevents unauthorized access.
- More secure than traditional keys.
- Tracks access attempts.

2. CCTV Cameras at Entrances and Hallways

Reasoning:

- Deters unauthorized activity.
- Provides evidence during investigations.
- Covers critical access points.

3. Security Badge Access at Main Entrance

Reasoning:

- Ensures only authorized personnel enter.
- Creates access logs for accountability.
- Strengthens perimeter security.

4. Electronic Key Card Access System at executive room

Reasoning:

- Ensures that only authorized employees can enter restricted areas.
- It also creates an audit trail showing who accessed specific locations and when.

5. Locked Server Room with Reinforced Doors

Reasoning:

- Protects critical servers and networking equipment from theft, tampering, or unauthorized access.
- Reinforced doors provide an additional layer of physical security.

6. Physical Isolation for AI Model Infrastructure (ISO 42001 / NIST AI RMF):

Reasoning:

- Prevents physical unauthorized tampering, side-channel attacks, or extraction of proprietary model weights directly from server memory or local storage.

STEP 3

Identify three environmental controls to implement on the first floor. These controls must minimize the loss of valuable data in the event of a fire, natural disaster, or power failure.

Explain your reasoning for each choice.

Environmental Controls:

1. Fire Suppression System (Clean Agent System)

Reasoning:

- Extinguishes fires without damaging electronic equipment.
- Protects servers and networking devices.

2. Uninterruptible Power Supply (UPS)

Reasoning:

- Provides backup power during outages.
- Prevents data corruption and unexpected shutdowns.

3. Climate Control System (HVAC)

Reasoning:

- Maintains safe temperature and humidity levels.
- Prevents overheating and hardware failure.

4. Backup Generator

Reasoning:

- Supplies electricity during prolonged power failures.

- ensuring continuous operation of critical systems and business processes.

5. Water Leak Detection Sensors

Reasoning:

- Detects leaks from pipes, roofs, or flooding before they cause significant damage to servers, cables, and other IT equipment.

6. Hardware Monitoring for High-Density AI Compute (implementing specialized liquid/coolant sensors)

Reasoning:

- AI training and inference hardware generates high thermal outputs; targeted cooling controls prevent thermal throttling and hardware degradation.

Conclusion

This project focused on identifying and implementing effective physical and environmental security controls to protect an organization's information systems, infrastructure, and confidential data. The assessment identified potential security risks and proposed measures to reduce vulnerabilities and strengthen overall security. Physical controls such as biometric access systems, CCTV surveillance, security personnel, electronic access cards, and secured server rooms help prevent unauthorized access and protect critical assets. Environmental controls including fire suppression systems, UPS devices, backup generators, HVAC systems, and water leak detection sensors help ensure business continuity and safeguard IT equipment from environmental hazards. By implementing these controls, the organization can improve the confidentiality, integrity, and availability of its information systems while reducing the likelihood and impact of security incidents.